

THEJUS MAHAJAN PH.D.

Bioinformatics & Data Scientist

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CAREER INTERESTS

Computational Scientist (PhD) with 5+ years of experience in **Biological Systems Modeling** and **High-Dimensional Data Analysis**. Experienced during post-doc in applying **stochastic population models** in hindcast and forecast projections. Trained in **Python and R pipelines** for noisy data analysis. Seeking for a career shift to Bioinformatics. I am leveraging on my recent Bioinformatics and Biostatistics training along with the advanced mathematical modeling expertise gained during the Ph.D. and post-doc period to solve problems in clinical data analysis. Currently in an internship phase in clinical data research and development which is my stepping stone for a full time professional career in the Bioinformatics sector in Germany.

EXPERIENCE

Internship - Clinical Biostatistics

HealthTwist GmbH

02/2026 - Present Berlin, Germany

Clinical data research and development

- Working on the development of R pipelines for quality reporting on 143,000+ interventional radiology procedures (GDPR-compliant medical data).

Further Training - Application-related Bioinformatics and Biostatistics

CQ Beratung + Bildung GmbH

08/2025 - 02/2026 Berlin, Germany

Further Training in Bioinformatics and Biostatistics

- Introduction to programming and Python, including Shell (BASH), programming methodology, and object-oriented programming in Python.
- Bioinformatics and databases, including NCBI, Biopython, web services, and database management with SQL.
- Applied bioinformatics and biostatistics, including sequence analysis, structural bioinformatic, NGS methods, statistical analysis using R, descriptive and multivariate statistics, ANOVA, PCA, HCA, workflows using R/Bioconductor, and Markov chain implementation.

Guest Scientist

Helmholtz-Zentrum Hereon, Ökosystemmodellierung

05/2025 - 10/2025 Geesthacht, Germany

Computational Biologist & Systems Modeler

- Spearheaded a complex research project modeling evolutionary processes in marine ecosystems, collaborating with Ecosystem Modeling Department.
- Applied advanced computational techniques to simulate the long-term impacts of environmental changes on marine ecosystems.

Post-doctoral Scientist

University of Hamburg, Institute of Marine Ecosystem and Fisheries Sciences

08/2021 - 01/2025 Hamburg, Germany

Marine Ecosystem Modelling

- Model development: Developed the novel "Cyanobacteria Life Cycle" (CLC) model within the ERGOM framework to project future climate warming scenarios.
- High-performance computing: Engineered models in Fortran and analyzed large NetCDF datasets using Python in a Linux environment.

STRENGTHS



Advanced Scientific Modelling

Developed and implemented a novel biogeochemical model (CLC) within the GOTM-FABM framework to run complex climate change projection scenarios.



Complex Problem-Solving

Resolved a critical data-loss error by creating a Fortran simulation to correct for a misplaced particle detector, successfully recalculating experimental results using multi-sensor data.



Bioinformatics & Biostatistics

Successfully bridged physical science expertise with modern biological workflows. Skilled in NGS data analysis, sequence alignment, and multivariate statistical modeling using R (Bioconductor) and Python.

TECHNICAL SKILLS

Programming

R, Python, Fortran, Nextflow, SQL, BASH

Bioinformatics Tools

NGS Analysis, Galaxy, Sequence Alignment, Structural Bioinformatics

Biostatistics

Multivariate Analysis, PCA, ANOVA, Dimensionality Reduction

LANGUAGES

English



Proficient

German



B1 - Goethe Certified

French



Intermediate

Malayalam



Native

Tamil



Proficient

Hindi



Intermediate

Job search

📅 04/2021 - 07/2021 📍 Hamburg, Germany

- Actively searched for a Ph.D. level job in and around Hamburg.

Physics Subject Matter Expert

Tutorwaves Solutions Inc

📅 08/2020 - 03/2021 📍 Kerala, India

E-Learning Providers

- Online Physics classes tailored to students appearing for competitive examinations like GCSE, AIEEE etc.
- Prepared Physics question banks for the purpose of mock examinations.

Physics Trainer and Chess Coach.

Freelancer

📅 10/2018 - 07/2020 📍 Kerala, India

Physics and Chess Educator

- Prepared student appearing for National level examinations like AIEEE, CBSE Board etc.
- Coached several chess teams towards winning their state titles.

Ph.D. Researcher in Astrochemistry

University of Paris-Saclay

📅 10/2015 - 09/2018 📍 Orsay, France

Led atomic collision experiments, analyzed data, and modeled outcomes.

- Big Data Analysis: Acquired and processed terabytes of raw data from tandem particle accelerator experiments.
- Algorithm Development: Utilized C++ for noise cancellation, signal smoothing, and converting raw data into multidimensional structures.
- Simulation: Developed a molecular fragmentation model to predict chemical reactions in space, contributing results to the KIDA database.

REFERENCES

Dr. Andreas Busjahn
abusjahn@healthtwist.de

Prof. Dr. Kai Wirtz
kai.wirtz@hereon.de

Prof. Dr. Elisa Schaum
elisa.schaum@uni-hamburg.de

TRAINING / COURSES

High Performance Computing (HPC) Training
Jülich Supercomputing Center (JSC),
Forschungszentrum Jülich

MITx: 7.00x Introduction to Biology - The Secret of Life

Comprehensive coursework on genetics, biochemistry, and molecular biology.

EDUCATION

Ph.D. in Astrochemistry

Université Paris-Saclay, Institute of Molecular Sciences (ISMO) Orsay, France

📅 09/2015 - 09/2018 📍 Orsay, France

Independent research

📅 02/2015 - 08/2015 📍 Calicut, India

M.Sc. in Physics

National Institute of Technology Calicut, India

📅 07/2012 - 01/2015 📍 Calicut, India

B.Sc. in Physics

University of Calicut, India

📅 06/2009 - 04/2012 📍 Thrissur, India

PUBLICATIONS

T. Mahajan, et al. *Journal of Physics: Conference Series*, 1412:142026, 2020.

T. Idbarkach, et al. *Astronomy & Astrophysics*, 628:A75, 2019.

T. Mahajan, et al. *Journal of Physics B*, 2019.

T. Idbarkach, et al. *Journal of Physics B*, 51(24):245201, 2018.

T. Idbarkach, et al. *Journal of Physics: Conference Series*, 1412(11):112028, 2020.